

HAIKAL HAKIM BAIQUNNI

R&D Engineer — Mechatronics × AI-Augmented Engineering × Legacy Modernization

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PROFILE

Mechatronics and mechanical engineer with 10+ years across precision measurement R&D, motor control design, and smart sensor development, currently at Mitutoyo Corporation HQ, Japan. Daily user of AI coding tools (Claude Code, OpenAI Codex, Google Antigravity, Cursor, GitHub Copilot) applied to real engineering work — accelerating a COMSOL/MATLAB/Python simulation pipeline from days to hours per iteration, refactoring legacy tooling, and building GUI and CAD software from scratch. Comfortable moving across C/C++, Python, and MATLAB, reading unfamiliar or undocumented code, and knowing when to override an AI suggestion that's wrong. Published IEEE & SICE author with a filed patent. Trilingual (Indonesian, English, Japanese — JLPT N3).

AI-AUGMENTED ENGINEERING — WORKFLOW TOOLS

Routinely applies AI coding and workflow tools to compress development cycles — from simulation scripting to documentation and rapid prototyping. Uses AI as an engineering multiplier, not a replacement for technical judgment.

Claude Code

Agentic CLI for end-to-end coding tasks — simulation scripts, data pipelines, documentation generation, iterative debugging across COMSOL/MATLAB/Python workflows

OpenAI Codex

Code generation and refactoring for Python evaluation tools, GUI development (PySide6), sensor data analysis pipelines

Google Antigravity

Multi-agent browser orchestration for cross-window task automation and development workflow management

Cursor

AI-native IDE for context-aware editing and codebase-wide refactoring across mechatronics tooling projects

GitHub Copilot

Inline assistance for boilerplate generation, test writing, and documentation in version-controlled projects

Applied at Mitutoyo: Used Claude Code and Codex to accelerate development of the COMSOL–MATLAB–Python simulation pipeline (ecs_generator_v5.py), cutting iteration time on parametric sweep scripts from days to hours. Also applied AI tooling to build a PySide6 PCB CAD editor and to generate technical documentation.

PROFESSIONAL EXPERIENCE

Research and Development Engineer — Mechatronics Development Section 2

Mitutoyo Corporation

Apr 2023 – Present

Kawasaki, Japan (HQ)

Focus 1: Lens / Optics Non-Contact Measurement

- Lead development of a non-contact measurement instrument for precision lens evaluation, integrating vision systems, electromagnetic actuation, and control algorithms.
- Developed WLO lens measurement system with dual SIM sensors, XY magnetic linear motor stage, and decenter calibration algorithm — achieving 0.8–1.0 μm accuracy across 3 delivered prototypes.
- Project resulted in filed patent DE102024129207A1 (Mitutoyo Corp, Oct 2024). International collaboration with Mitutoyo RCE Netherlands & SIOUX Technologies.

Focus 2: Smart Eddy Current Sensor (ECS) — New Business Development

- Selected member of Mitutoyo New Business Strategic Development Office, driving a smart sensor product from ideation through feasibility and prototype.
- Built end-to-end EM simulation pipeline: COMSOL 6.4 → MATLAB LiveLink → Python GUI (ecs_generator_v5.py), with development accelerated by AI coding tools (Claude Code, Codex) — including refactoring undocumented legacy scripts into a maintained tool with a proper Results/Plot module.
- Delivered a functional planar ECS prototype for gap/thread surface evaluation. Joint R&D with Mitutoyo Research & Development America (MRDA).

Technical Support Supervisor

PT. Mimaki Indonesia

Aug – Nov 2019

Jakarta, Indonesia

- Supervised technical support team for printer installation and field maintenance across Indonesia.
- Primary Japanese–Indonesian–English interpreter between Japanese management and local teams.

Printer Designer — Mechanical Design & Motor Control Engineer

PT. Indonesia Epson Industry · Seiko Epson Corporation (Secondment)

Dec 2013 – Mar 2019

Indonesia / Japan

Mechanical Design Engineering

- Designed printer mechanism components and full mechanical systems for inkjet product lines — 30% efficiency improvement through subsystem redesigns.
- Led DFM evaluations, ECOs, and design documentation through formal PLM workflows.

Motor Control Design Engineering

- Designed PID control parameters and motor control sequences for BIJ-model inkjet printers across paper feed, carriage, and media handling systems — working in a long-lived, partially legacy C/C++ codebase requiring careful incremental changes without disrupting existing production behavior.
- Tuned DC motor, stepper motor, and carriage control systems across FY15–FY18 product generations.

Seiko Epson Corp. Secondment — Hirooka Office, Japan (May 2014 – May 2017)

- Selected for 3-year Japan HQ secondment; contributed to FY15–FY18 model development and received formal design team leadership training.
- New Motor Adaptation project: led DC motor spec transition (Mabuchi RS385PH/445PH → RS385PV/RS425PV) — delivered USD \$100,000 cost reduction.

EDUCATION

M.Eng, Mechanical System Engineering and System Design — Tokyo Metropolitan University

MEXT Scholarship Recipient · 2021 – 2023 · Thesis: Sliding mode control for drone attitude stabilization (published, SICE 2023)

B.Eng, Mechatronics Engineering — Institut Teknologi Sepuluh Nopember (ITS), Surabaya · 2009 – 2013

TECHNICAL SKILLS

Engineering	Mechatronics design, motor control (PID, sequences), non-contact measurement, vision systems, sliding mode control, electromagnetic actuation, UAV/quadcopter control
Software	Python, C/C++, MATLAB/Simulink, COMSOL 6.4, neural networks / AI, legacy code refactoring & modernization
AI Dev Tools	Claude Code, OpenAI Codex, Google Antigravity, Cursor, GitHub Copilot
PCB Design	Altium, KiCAD, EasyEDA — custom sensor PCB design (ECS planar & stacking type)
CAD / Mfg	SolidWorks, Autodesk Fusion 360, PTC Creo, 3D printing, DFM/DFA
Languages	Indonesian (Native) · English (EF SET C1 Advanced) · Japanese (JLPT N3)

PUBLICATIONS & PATENT

- Haikal Hakim Baiqunni et al. A Trajectory Generation Design for a Sliding Mode Control of a Quadcopter for Chattering Attenuation. SICE Annual Conference 2023, Mie University, Japan.
- Haikal Hakim Baiqunni et al. Vowel Recognition System of Lipsynchrobot in Lips Gesture Using Neural Network. 2014 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE).
- Nakajima Y. et al. (2024). Measuring Machine — dual-sensor WLO measurement system. Patent No. DE102024129207A1. Filed Oct 2024, Published Apr 2025. Assignee: Mitutoyo Corporation. Status: Pending.